Customer Shopping Behaviour Analysis

1. Project Overview

This project analyzes customer shopping behaviour using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behaviour to guide strategic business decisions.

2. Dataset Summary

Rows: 3,900 - Columns: 18 - Key Features:
Customer demographics (Age, Gender, Location, Subscription Status)
Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Colour)
Shopping behaviour (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
Missing Data: 57 values in Age, Gender & Location columns; 96 in Review Rating; 59 in Payment Method

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
count	3900.000000	3843.000000	3843	3843	3900	3900.000000	3843	3900	3900	3900	3804.000000	3900	3900	3900	3900	3900.000000	3841	3900
unique	NaN	NaN	7	27	4	NaN	100	4	25	4	NaN	2	6	2	2	NaN	33	7
top	NaN	NaN	Male	Jacket	Clothing	NaN	Nagpur	M	Olive	Spring	NaN	No	Free Shipping	No	No	NaN	Paytm	Every 3 Months
freq	NaN	NaN	1565	322	1737	NaN	102	1754	176	999	NaN	2846	677	2223	2223	NaN	361	585
mean	1950.847179	44.033568	NaN	NaN	NaN	5497.099487	NaN	NaN	NaN	NaN	3.748922	NaN	NaN	NaN	NaN	25.379744	NaN	NaN
std	1127.147545	15.211922	NaN	NaN	NaN	31920.096040	NaN	NaN	NaN	NaN	0.717155	NaN	NaN	NaN	NaN	14.455480	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	-500.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN	1.000000	NaN	NaN
25%	973.750000	31.000000	NaN	NaN	NaN	2882.500000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN	13.000000	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	4507.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN	25.000000	NaN	NaN
75%	2926.500000	57.000000	NaN	NaN	NaN	6054.250000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN	38.000000	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	999999.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN	50.000000	NaN	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the `Review Rating` column using the median rating of each product category.

- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.

- **Feature Engineering:**

  - Created **age_group** column by binning customer ages.

  - Created **purchase_frequency_days** column from purchase data.

- **Data Consistency Check:** Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.

- **Database Integration:** Connected Python script to MySQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in MySQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	gender	revenue
0	Female	8606539.0
1	Male	12832149.0

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id	purchase_amount
0	4	6894
1	7	6520
2	9	7344
3	16	5946
4	20	6904
...	...	...
544	1656	6013
545	1662	6467
546	1676	6873
547	957	7301
548	889	5674

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

	item_purchased	Average Product Rating
0	Shoes	3.89
1	Kurta	3.87
2	Sneakers	3.83
3	Sandals	3.81
4	Cap	3.80

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

	shipping_type	avg_purchase_amount
0	Express	4539.56
1	Standard	4369.39

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	subscription_status	total_customers	avg_spend	total_revenue
0	No	2846	5534.21	15750367.0
1	Yes	1054	5396.89	5688321.0

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased	discount_rate
0	Sunglasses	49.60
1	Coat	49.07
2	Loafers	47.73
3	Belt	47.54
4	Mojaris	47.50

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment	Number_of_Customers
0	Loyal	3116
1	New	84
2	Returning	700

8. **Top Products per Category** – Listed the most purchased products within each category.

	category	item_purchased	total_orders
0	Accessories	Backpack	161
1	Accessories	Shawl	154
2	Accessories	Scarf	153
3	Accessories	Wallet	136
4	Accessories	Watch	134
5	Accessories	Sunglasses	125
6	Accessories	Belt	122
7	Accessories	Cap	120
8	Accessories	Handbag	114
9	Accessories	Jacket	17
10	Clothing	Jacket	190
11	Clothing	Kurta	186
12	Clothing	T-Shirt	183
13	Clothing	Trousers	179
14	Clothing	Shirt	176
15	Clothing	Kurti	174
16	Clothing	Ethnic Top	172
17	Clothing	Jeans	164
18	Clothing	Saree	157
19	Clothing	Lehenga	156
20	Footwear	Slippers	93
21	Footwear	Sneakers	89
22	Footwear	Loafers	88
23	Footwear	Shoes	85
24	Footwear	Boots	82
25	Footwear	Mojaris	80
26	Footwear	Sandals	77
27	Footwear	Jacket	7
28	Outerwear	Jacket	165
29	Outerwear	Coat	161

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

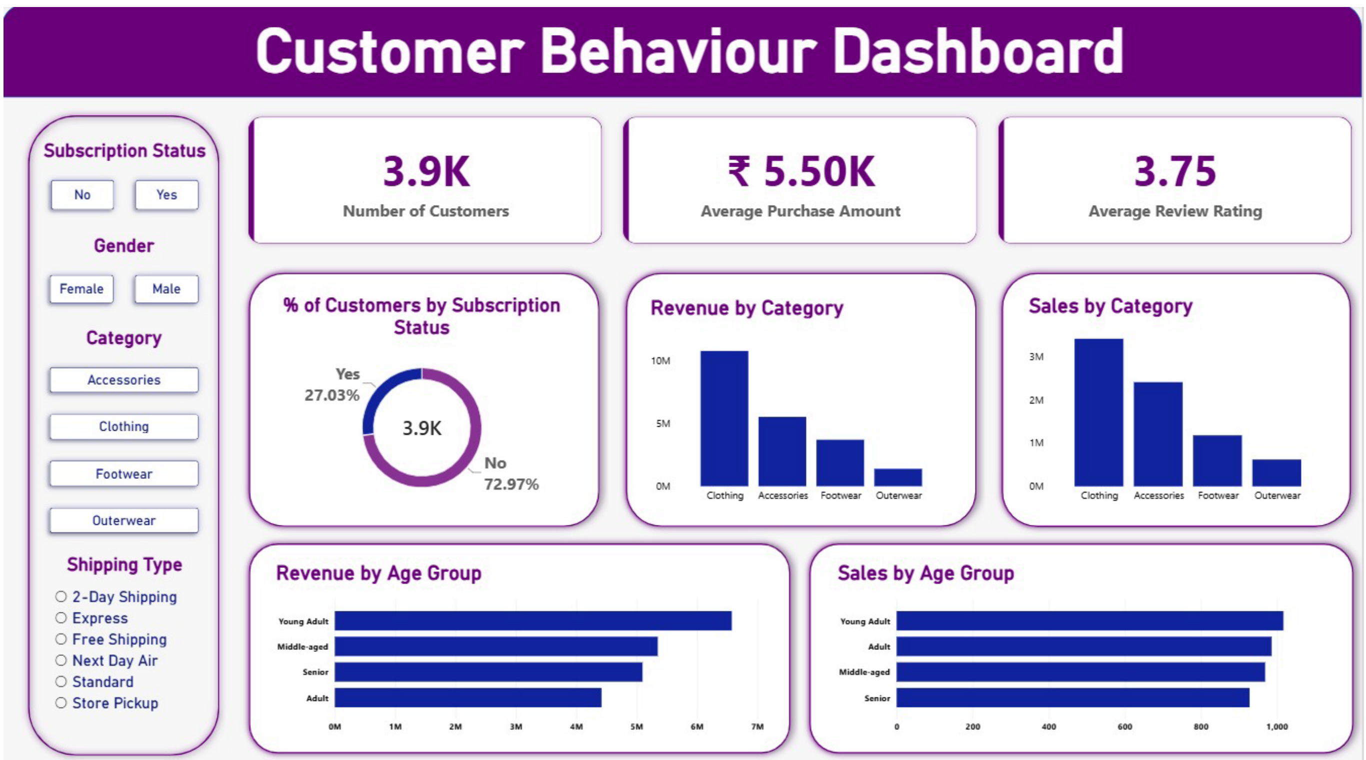
	subscription_status	repeat_buyers
0	No	2518
1	Yes	959

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

	age_group	total_revenue
0	Under 25	6574915.0
1	40-59	5349233.0
2	60+	5096600.0
3	40-59	4417940.0

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



6. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.